Durga Chandhan
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SUMMARY

- **Data Engineer** with **5+ years of experience** designing, developing, and optimizing large-scale data pipelines and cloud-based data architectures across **Microsoft, NVIDIA,** and **Abbott**. Proven expertise in building end-to-end **ETL/ELT workflows**, **real-time streaming solutions**, and **machine learning pipelines** using technologies such as **Azure** (ADF, Databricks, Synapse), **AWS** (SageMaker, Glue, Redshift), **Apache Spark**, **Kafka**, **Airflow**, and **Delta Lake**.
- Skilled in managing multi-terabyte datasets, integrating structured and **unstructured data from IoT**, clinical, and enterprise systems. Strong background in **Python (PySpark, Pandas, scikit-learn)**, **SQL**, and **DevOps tools** including **Jenkins**, **Docker**, **Kubernetes**, and **Terraform** for **CI/CD automation**.
- Recognized for driving data accuracy improvements, cost optimizations, and **ML-driven automation**, with measurable impact such as reducing **ETL costs** by **45%**, anomaly resolution time by **60%**, and **AI data** errors by **35%**. Adept at cross-functional collaboration, mentoring, and delivering actionable insights through **Power BI**, **DAX**, and real-time dashboards for enterprise-wide stakeholders.

SKILLS

Programming & Scripting: Python (Scikit-learn, TensorFlow, PyTorch, Keras, NLTK, PyMC3), Scala, SQL, PL/SQL, Shell Scripting
Cloud Platforms: Azure (ADF, Databricks, Synapse, Blob Storage, Data Lake), AWS (EC2, S3, Redshift, DynamoDB), GCP (BigQuery)
Big Data & Processing Frameworks: Apache Spark, Apache Hadoop, Apache Hive, HBase, MapReduce, Amazon EMR, Delta Lake
Data Engineering & Pipeline Tools: ADF, Azure Databricks, Apache Airflow, Apache Kafka, Kinesis, dbt
Databases & Data tools: Microsoft SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, Snowflake, Redshift, DynamoDB, TimescaleDB
Machine Learning & AI: Feature Engineering, Predictive Maintenance Models, Model Training Pipelines, Large Language Models (LLMs), Azure OpenAI, LangChain, Natural Language Processing (NLP), Anomaly Detection, Time-Series Analysis
DevOps & Automation: Git, Jenkins, Docker, Kubernetes, GitHub Actions, ArgoCD, Terraform, Bicep, Azure DevOps, CI/CD Pipelines
Data Visualization & Reporting: Power BI, Tableau, QlikView, Microsoft Excel, SSRS, DAX Calculations, DirectQuery Connections
Project Management & Methodologies: Agile, SDLC, Jirax

CERTIFICATIONS

- **Microsoft Certified:** Fabric Analytics Engineer Associate (**DP600**), Power BI Data Analyst Associate (**PL300**).
- **Databricks Certified:** Data Engineer Associate.

EXPERIENCE

Microsoft | AI & ML Data Engineer — **WA | June 2024 – Present**
- Designed and deployed **12+ scalable data pipelines** using **Azure Data Factory**, **Azure Synapse Analytics**, **Azure Databricks**, and **Microsoft Fabric**, leveraging **Delta Lake Medallion architecture** to reduce ETL costs by **45%** and improve data freshness.
- Partnered with **AI** and **product** teams to design and deploy **end-to-end data pipelines** on **Azure** (Data Factory, Synapse, and Databricks) powering real-time **ML models** for personalized user insights; leveraged Prompt Flow and **Azure OpenAI** to operationalize **LLM-based** analytics, **improving recommendation** accuracy by **30%**.
- Created **machine learning pipelines** in **Python**, **PySpark**, and **scikit-learn** on **Databricks**, automating **feature engineering** for **predictive maintenance models** and improving downtime prediction accuracy to **92%**, reducing prediction window from **4 hours to 15 minutes**.
- Built **LLM-powered data quality monitoring** solutions using **Azure OpenAI**, **LangChain**, and automated **SQL query generation**, reducing data anomaly resolution time by **60%** across **30+ datasets**.
- Provided mentorship to **2 junior data engineers**, offering guidance in **Azure architecture**, **PySpark**, and **dbt best practices**, leading to a **40% increase** in project contribution time and team-wide adoption of standardized **data transformation patterns**.

NVIDIA | Data Engineer — **CA | September 2023 – June 2024**
- Built and scaled **machine learning pipelines** on **AWS** leveraging **SageMaker**, **EC2**, and **S3**, increasing data processing efficiency by **50%** and enabling **real-time model deployment**.
- Optimized **quantitative data pipelines** using **CUDA** and **NVIDIA RAPIDS**, enhancing data access speed by **10%**; engineered **real-time feature pipelines** using **MongoDB** and **NVIDIA Omniverse** for dynamic ML insight visualization, reducing time-to-decision by **5%**.
- Developed robust **data validation scripts** in **Python** (**PySpark**, **Pandas**) and **SQL**, integrating **Kafka** (streaming) and **Delta Lake** (batch) data sources with **window functions** and **UDFs** to reduce **training data anomalies by 35%**.
- Implemented **CI/CD pipelines** using **Jenkins** and **Kubernetes Operators**, automating testing and deployment of **20+ data pipelines**, shrinking release cycles from **3 days to 4 hours** with **99.8% reliability**.
- Automated data preprocessing and vector embedding generation for **NLP workloads** using **Hugging Face Transformers**, **FAISS**, and **PyTorch**, enabling **RAG** (**Retrieval-Augmented Generation**) **pipelines** with real-time document search.

Abbott | Data Engineer — **India | June 2019 – August 2022**
- Implemented a scalable **Snowflake data warehouse** with **optimized SQL queries** and **stored procedures**, centralizing clinical trial data from **50+ sources** and reducing query execution time for analytics by **60%**.
- Built and maintained **Hadoop data lake architectures** using **HDFS**, **Hive**, and **Apache Spark**, enabling daily processing of over **10TB** of clinical trial data; improved **ETL runtimes by 35%** through optimized **MapReduce**, while ensuring **HIPAA** and **GDPR** compliance.
- Designed **real-time data ingestion pipelines** using **Apache Kafka** and orchestrated workflows via **Apache Airflow**, processing **5K+ patient vitals per second** from **IoT medical devices** and automating **data validation**, improving data availability for researchers by **50%**.
- Created **Python scripts** using **Pandas** and **NumPy** for **data cleansing**, normalization, and transformation of clinical trial datasets, enhancing data quality by **20%** and accelerating research cycles.

EDUCATION

Master's in Computer Science | Rivier University, Nashua, New Hampshire, USA.